

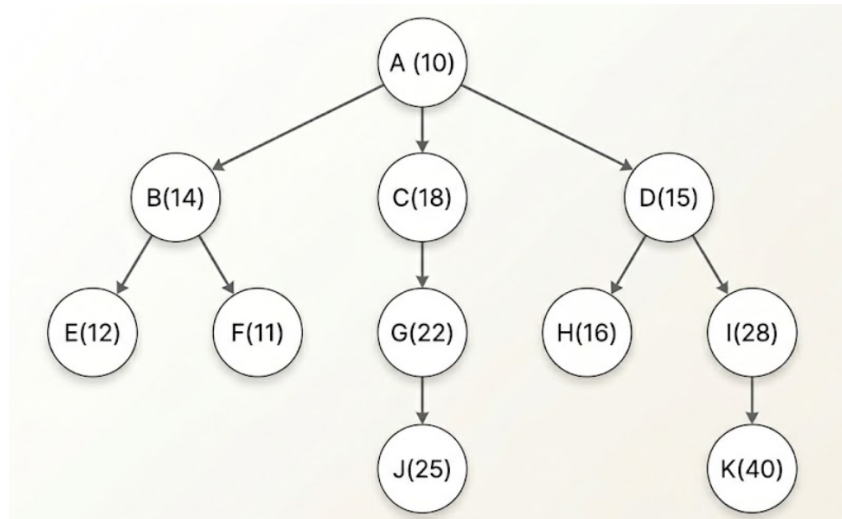
FAST NUCES ISLAMABAD CAMPUS

Class Activity - Local Search

Section B/E/F

Student Name: _____ Roll Number: _____

State Space Diagram



PART A — Hill Climbing

1. Perform a **Hill Climbing** search starting from Node **A**.
2. List the sequence of nodes visited.
3. Does this algorithm reach the Global Maximum **K(40)**? Explain why it stops where it does.

PART B — Local Beam Search

- **Use:** Beam width $k = 2$
- **Tasks:**
 1. Run Local Beam Search step-by-step starting from **C**.
 2. Show the **successor pool** at each iteration.
 3. Identify which branches are discarded at each step.
 4. Does this approach successfully reach **K(25)**?

PART C — Simulated Annealing

- **Use:** Temperature schedule: $T_0 = 10$; Cooling: $T = T - 2$ each step.
- **Tasks:**
 1. Run Simulated Annealing step-by-step starting from **A**.

Questions:

1. Explain the probability of moving to a state with a lower value when the temperature T is high versus when T is low.
2. Why is Simulated Annealing more likely to reach the Global Maximum than standard Hill Climbing in this specific tree?

— *End* —